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Original Article

Development and pilot feasibility of Kit-CAT, a serious-game-supported consultation for discussing complementary and alternative therapies in breast cancer care

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ABSTRACT

Background: Patients with early breast cancer frequently consider complementary therapies, yet structured, ethical discussions remain uncommon. Breast cancer is a particularly relevant context for such discussions because care pathways are prolonged, multimodal, and often associated with uncertainty, treatment-related symptoms, and high informational burden.
Aim: We developed **and pilot tested** Kit-CAT, a serious-game-supported consultation to scaffold evidence-based, patient-centered conversations on complementary therapies.
Methods: After developing Kit-CAT through an iterative process including clinicians, researchers, and patients, we conducted a monocentric before-and-after pilot (6-month periods). Women with non-metastatic breast cancer were included at diagnosis; the intervention added a 90-minute primary-care consultation. Feasibility was assessed across recruitment, delivery, and acceptability. No formal a priori threshold for feasibility success was specified. Outcomes included feasibility indicators, health literacy (HLQ), attitudes to CAT (CHBQ), CAT use, and satisfaction at baseline and 6 months.
Results: Of the 44 initially included participants, 41 were retained in the baseline analysis set and 38 provided at least one 6-month follow-up data point (25 in the control group and 13 in the intervention group). Recruitment to the intervention proved challenging: during the intervention phase, 49 eligible patients declined participation. Recruitment feasibility was therefore limited under the tested implementation conditions. Across several HLQ subscales, small descriptive trends in a favourable direction were observed in the intervention group, contrasting with modest declines in the control group; attitudes toward CAT were highly positive at baseline and remained stable. Satisfaction with the care pathway was high in both groups, with the intervention group numerically higher (92%vs 78% satisfied/very satisfied). No serious adverse events were reported; recorded adverse events were not considered related to Kit-CAT.
Conclusion: Kit-CAT was deliverable and acceptable among consenting participants, but recruitment feasibility was limited, mainly because of early timing after diagnosis, off-site delivery, and scheduling burden. Descriptive health-literacy trends should be interpreted as exploratory. Future multicentre studies should first optimize implementation format before assessing effectiveness in an adequately powered design.

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1. Introduction

Breast cancer is the most common cancer among women in France, accounting for over 30% of female cancers [1]. This highly prevalent pathology is extremely heterogeneous. Depending on the stage at diagnosis, the characteristics of the tumour, and the risk factors for recurrence, different treatments may be proposed. The management can be more or less complex and may include surgery (lumpectomy or mastectomy), chemotherapy, neoadjuvant and/or adjuvant radiotherapy, hormone therapy, and/or immunotherapy.

Its management often involves a prolonged, multimodal care pathway combining surgery, radiotherapy, systemic treatments, endocrine therapy, surveillance, and supportive care. Throughout this pathway, women may experience fatigue, pain, anxiety, menopausal symptoms, sleep disturbances, treatment-related adverse effects, and uncertainty regarding prognosis and recurrence. These experiences may lead patients to seek complementary and alternative therapies (CAT) not only as symptom-management strategies, but also as ways of regaining agency and control during a highly medicalized trajectory. In this context, the issue is not only whether CAT are effective or ineffective, but whether patients can discuss them safely, openly, and in an evidence-informed manner with healthcare professionals.

The use of complementary therapies has been documented since the 1970s [2]. The results of a recent systematic review indicate that more than half of these patients utilize CAT [3]. A global survey has also revealed that this proportion has been steadily increasing since the 1990s [4]. Aligning with the population affected by breast cancer, the profile associated with high CAT use generally corresponds to a young woman with both a high level of education and socioeconomic status [3]. Although some evidence on CAT communication comes from oncology more broadly, these issues are especially salient in breast cancer because of the high prevalence of CAT use, the long duration of treatment and follow-up, and the potential for interactions with systemic therapies, including endocrine and anticancer treatments.

Although this use of CAT is common among cancer patients, it remains controversial within the medical and scientific communities [5]. While their use may help relieve disease symptoms or treatment side effects, it can also be harmful if used without supervision [6,7]. CATs are often criticized for being based on weak theoretical foundations, for lacking evidence of effectiveness beyond a placebo response in methodologically sound studies, and for posing risks to patients. Among these risks, the potential for interactions between conventional cancer treatments and CAT is significant [8,9]. While these criticisms are generally justified, these controversies also contribute to increasing the anxiety levels of patients who are considering using these therapies as part of their cancer management or to cope with its effects [10,11]. In China, cancer care is a health event that triggers an initial consultation in CAT for 77% of female patients [12]. Moreover, it has been observed that patients do not spontaneously discuss CAT with their doctors due to fear of their reaction [13,14]. When they do, it is most often raised at their initiative, despite it being the responsibility of healthcare providers [15].

Beyond these healthcare considerations, it is necessary to restore the caregiver-patient relationship within the currently favoured approach, patient-centered care. Consultations that follow a communication model consistent with this care philosophy result in better communication about CAT and greater satisfaction among both patients and caregivers [16,17]. This is partly explained by the fact that patients using CAT often have expectations that do not align with the actual outcomes that can be expected. Some patients, for example, expect a cure and end up disappointed after their consultation [18]. This is all the more relevant given that the relationship between the patient and the healthcare provider is a key determinant of treatment success. It influences several factors, such as adherence and self-care skills, which in turn impact health and well-being [19].

While this communication model is generally associated with higher quality of care, it requires adherence to a certain ethics of care, special attention to patients' preferences, and support in helping them make informed health decisions. It has become clear that decision-making regarding CAT by cancer patients is a complex, dynamic, non-linear, and highly individualized process. The need for support in this process has been widely reported in the literature [20]. One effective way to convey complex information to patients about their care pathways and promote agency in their decision-making is with a serious game [21]. In breast cancer care, such a tool may be particularly useful because decisions about CAT often occur at the intersection of symptom burden, treatment uncertainty, fear of recurrence, and exposure to heterogeneous sources of information. To the best of our knowledge, no such game has yet been developed and tested to inform about CAT in the context of breast cancer. Therefore, this work aims to design, implement, and pilot test the feasibility and exploratory outcomes of a specific consultation using a serious game on the role of CAT in the care pathway of patients with breast cancer.

2. Methods

2.1. Study overview

This was an intervention development and implementation study of a serious-game-structured consultation addressing CAT, with an embedded pragmatic before-and-after pilot in routine breast cancer care. The serious game serves as a facilitated conversation aid to support expectation setting, benefit-risk appraisal (including interaction warnings), and preparation for CAT discussions with the oncology team. The study was designed as a pilot feasibility study rather than as an effectiveness trial. The TiDieR Reporting guideline was used to describe the intervention development [22], and the TREND reporting guideline was used for the pilot study [23].

2.2. Intervention development

A multidisciplinary team of clinicians and researchers - comprising physicians, physiotherapists, and a PhD in science didactics - co-developed the intervention, with patients involved in an advisory role throughout. The needs assessment drew on the literature [20], reports from the French National Cancer Institute, and informal interviews with the oncology team (comprising oncologists, nurses, physiotherapists, and advanced practitioner nurses), as well as with patients. This confirmed the need to help healthcare providers open and structure CAT discussions in oncology and to answer common patient questions.

The serious game, Kit-CAT, was conceived as a non-prescriptive conversation aid within a shared-decision, patient-centred care model. Its objectives are to facilitate discussions and empower patients to make informed decisions and seek guidance. Kit-CAT was designed as a facilitated serious-game-supported consultation rather than as a standalone educational game. Its purpose is not to recommend or discourage specific CAT, but to create a structured, non-judgmental space in which patients can formulate questions, clarify expectations, identify possible benefits and risks, and prepare discussions with their oncology team. Therefore, to support truly informed consent about CAT, the game focuses on defining CAT, distinguishing between specific and non-specific effects, considering benefit-to-risk ratios, and fostering critical thinking about CAT in oncology. Anticipated benefits include improved health literacy and greater satisfaction with care and CAT use, resulting from more explicit expectation setting.

Its content is organised into three modules. The first module (1) Definitions and labels, covers how CAT is defined and how labels can hinder decisions, encouraging questions about the specific treatment under consideration. It helps distinguish between broad labels such as "natural", "alternative", "complementary", or "supportive" and the concrete practices to which these labels refer. The second module (2) Benefits and risks cover how benefits and harms are weighed and how

decisions regarding treatment acceptability considering benefit-risk are personal. It discusses benefits broadly and risks including potential harms, uncertainties, financial costs, delays in conventional treatment, and possible interactions with cancer therapies. Lastly, the third module (3) Trust and evidence covers how to judge health claims based on what is said, by whom, and where. It allows discussion on how to find reliable information in healthcare. This aims to help identify misleading arguments. The final artefact comprises 110 printed cards (46, 22, and 42 by module) released under a Creative Commons license, a facilitator guide, and a support website (https://www.timc.fr/jeu-ethic-tac).

During the consultation, the facilitator invites the patient to select, discuss, and prioritize cards according to her concerns. The cards are used to support dialogue rather than to impose a predefined route. The facilitator role is to encourage expression, clarify misunderstandings, provide evidence-informed explanations when appropriate, and help the patient identify questions to raise with oncology professionals. The consultation ends with a synthesis of the main issues discussed and, when relevant, preparation of questions for the oncology team.

The intended mechanisms of action are: improving access to understandable information; supporting critical appraisal of CAT-related claims; reducing fear of discussing CAT with healthcare professionals; making expectations more explicit; and improving the patient ability to navigate decisions involving CAT during breast cancer care. The structure of Kit-CAT is summarized in Table 1.

Regarding the development of Kit-CAT, prototyping and iteration involved successive versions created by four authors (LD, RM, NP, PG), which were tested with healthcare students, the research team, the oncology team, and ultimately with oncology patients. Feedback was discussed among the four creators and informed the development of new prototypes, including simplified wording, adjustments to certain cards, and addition of cards relevant to key needs identified by the partner oncology care team.

Before any deployment, a 3-hour training for physicians, delivered by the creators, covered the game and its rationale, the required posture (open, non-judgmental, non-prescriptive), basic CAT in oncology, guided practice with debrief, and a minute-by-minute guide (Supplementary) to support time management. Variations of the rules were discussed to encourage adaptation and to position the game as a discussion aid rather than a strict rule set. To support consistency of delivery, the facilitator guide provided a suggested sequence for the 90-minute consultation, including opening the discussion, selecting cards, exploring expectations, discussing benefits and risks, identifying reliable information sources, and preparing questions for the oncology team.

2.3. Implementation and pilot testing

Study design: We conducted a prospective, open-label, monocentric, quasi-experimental before-and-after study. Two consecutive 6-month periods were defined: a Before period (control) with usual care only and an After period (intervention) in which the serious game supplemented usual care. All participants were followed for 6 months after inclusion.

Table 1
Core components of Kit-CAT and intended functions.

Component	Description	Intended function
Module 1: Definitions and labels	Cards on terms such as natural, alternative, complementary, and supportive	Clarify vocabulary and avoid misleading labels
Module 2: Benefits and risks	Cards on expected benefits and harms including costs, delays, and interactions	Support benefit-risk appraisal
Module 3: Trust and evidence	Cards on information, people and sources	Strengthen critical appraisal and information literacy
Facilitator guide	Suggested consultation sequence, variations for the use of the cards and non-judgmental posture	Standardize delivery while preserving patient-centred adaptation

Setting and participants: Participants were recruited at the Women Health Department of Grenoble University Hospital. The intervention consultation was delivered at partner primary-care sites by physicians who had received training on the game. Inclusion criteria were women recently diagnosed ($\leq$3 months) with non-metastatic breast cancer (in situ or invasive; locally advanced without metastases confirmed histology), ECOG 0–2, undergoing chemotherapy and/or radiotherapy and/or surgery, adult, insured, and able to provide consent. Exclusion criteria were pregnancy/breastfeeding status, not French-speaking, enrolment in another study, metastatic breast cancer, psychiatric disorders preventing participation, prior cancers except cervical in situ or basal cell carcinoma, administrative/judicial supervision, or inability to be contacted in an emergency.

Assignment: Allocation followed calendar time (non-randomised before-and-after). To limit contamination, intervention consultations were delivered off-site; staffing, usual care pathways, and information materials remained unchanged across periods. To avoid disappointment bias, two separate information letters were used to present the study: one for the control (Before) group, mentioning participation included usual care and questionnaire responses, and another for the intervention (After) group, mentioning participation included usual care, questionnaire responses, and an additional consultation with a general practitioner.

The control (Before) period was conducted over two periods: from September 2022 to January 2023 and from June 2023 to August 2023. The intervention (After) period took place from November 2023 to May 2024. The initial recruitment of the control group was not continuous over six months; instead, it occurred over four months, followed by two months. The availability of clinical staff during the initial recruitment phase of the control group coincided with Breast Cancer Awareness Month, a particularly busy period for clinical staff, during which implementing a new protocol was challenging. This interval between the two control recruitment phases proved an opportunity to update and refine our recruitment procedure, expanding the number of people habilitated to screen for eligibility. These features are important for interpretation because they may have introduced period effects and differences in recruitment processes between groups.

Procedures: At D0 (inclusion), consent was obtained at the oncology clinic, and baseline questionnaires were completed. For the Intervention period, a research assistant scheduled a one-to-one consultation (90 min) with general practice physicians within the next two weeks. At M + 6 (end of study), follow-up questionnaires were collected by a research assistant over the phone. A nested qualitative study was carried out with a subset of patients [24].

Outcomes: The feasibility outcomes were recruitment capacity (number included over 6 months). Feasibility was assessed across three dimensions: recruitment feasibility, delivery feasibility, and acceptability. Recruitment feasibility was the primary feasibility outcome and was defined as the number of eligible participants who consented to participate during the 6-month intervention period, together with the reasons for refusal when spontaneously provided. Delivery feasibility was assessed by the ability to schedule and complete the consultation among consenting participants. Acceptability was assessed through satisfaction with the intervention and willingness to recommend it. No formal a priori threshold for feasibility success was defined, which is acknowledged as a limitation of this pilot study.

Other outcomes included: (i) care-pathway outcomes - Health Literacy Questionnaire (HLQ) [25]; a 4-level item (0–3) on satisfaction with improvement in navigating care; a yes/no item on capacity to navigate; and a 6-item bilateral Likert scale (−3 to +3; “very dissatisfied”/“very satisfied”); (ii) CAT-related outcomes - Complementary and Alternative Medicine Beliefs Questionnaire (CHBQ, patient version) [26]; an adapted simplified CAMP-Q [27]; and a 6-item bilateral Likert scale (−3 to +3; “strongly disagree” to “strongly agree”) on satisfaction with CAT only if CAMP-Q responses at 6 months were positive; and (iii)

oncology-care satisfaction - a 6-item bilateral Likert scale (−3 to +3). Denominators were reported for each available outcome because item-level missingness varied across questionnaires.

Sample size and analysis: This pilot was designed for descriptive analysis without formal hypothesis testing or power calculation. Quantitative analyses include descriptive statistics (counts, percentages, means, standard deviations, medians, and quartiles). No formal hypothesis testing was planned or performed. Descriptive statistics are presented with denominators for each outcome. Given the small sample size and variable denominators, confidence intervals were not used for inferential interpretation. As such, 95% CI for group differences would not be validly interpretable; however, 95% CI estimated for group scores are provided in Supplementary Materials.

Masking: The study was an open-label study with self-reported outcomes. Period-specific information leaflets were used, and participants were not informed about the existence of an alternative period to reduce disappointment bias.

Ethics and registration: All participants provided written informed consent. The study received approval from a French national ethics committee and was conducted in accordance with the GDPR and the Declaration of Helsinki. It was registered under ID number RCB 2021-A03027–34. The trial was prospectively registered on ClinicalTrials.gov (NCT05388734).

3. Results

The complete results report is available in Supplementary Materials.

3.1. Participants

Overall, 44 participants were initially included: 30 during the control period and 14 during the intervention period. In the control group, three participants were subsequently excluded before baseline because they had been erroneously included: one did not have a confirmed cancer diagnosis and two had a diagnosis older than permitted by the inclusion criteria. The baseline analysis set therefore comprised 41 participants: 27 in the control group and 14 in the intervention group. During follow-up, two additional control participants and one intervention participant were excluded either because they withdrew consent or were lost to follow-up. Thus, 38 participants had at least one 6-month follow-up data point: 25 in the control group and 13 in the intervention group, as described in Fig. 1. Baseline characteristics of participants are presented in Table 2.

Quantitative variables are expressed with mean (SD) and qualitative variables are presented with n (%). Baseline characteristics should be interpreted descriptively because of the small sample size and non-randomized before-and-after design. The two groups were broadly similar in age and comorbidity status, although the intervention group included a higher proportion of salaried participants and a lower

Table 2
Participant baseline characteristics.

Characteristic	Control group (n = 27)	Intervention group (n = 14)
Age, mean (SD)	60.8 (10.9)	58.6 (10.1)
Other chronic illnesses, n (%)	15 (55.6%)	7 (50.0%)
Socio-demographic categories		
Salaried	7 (25.9%)	7 (50.0%)
Executive/manager ("Cadre")	3 (11.1%)	2 (14.3%)
Intermediate occupation	3 (11.1%)	1 (7.1%)
Self-employed	1 (3.7%)	0
Unemployed or not working	2 (7.4%)	0
Retired	11 (40.7%)	4 (28.6%)
Affected breast		
Left	14 (51.9%)	8 (57.1%)
Right	11 (40.7%)	5 (35.7%)
Both	2 (7.4%)	1 (7.1%)

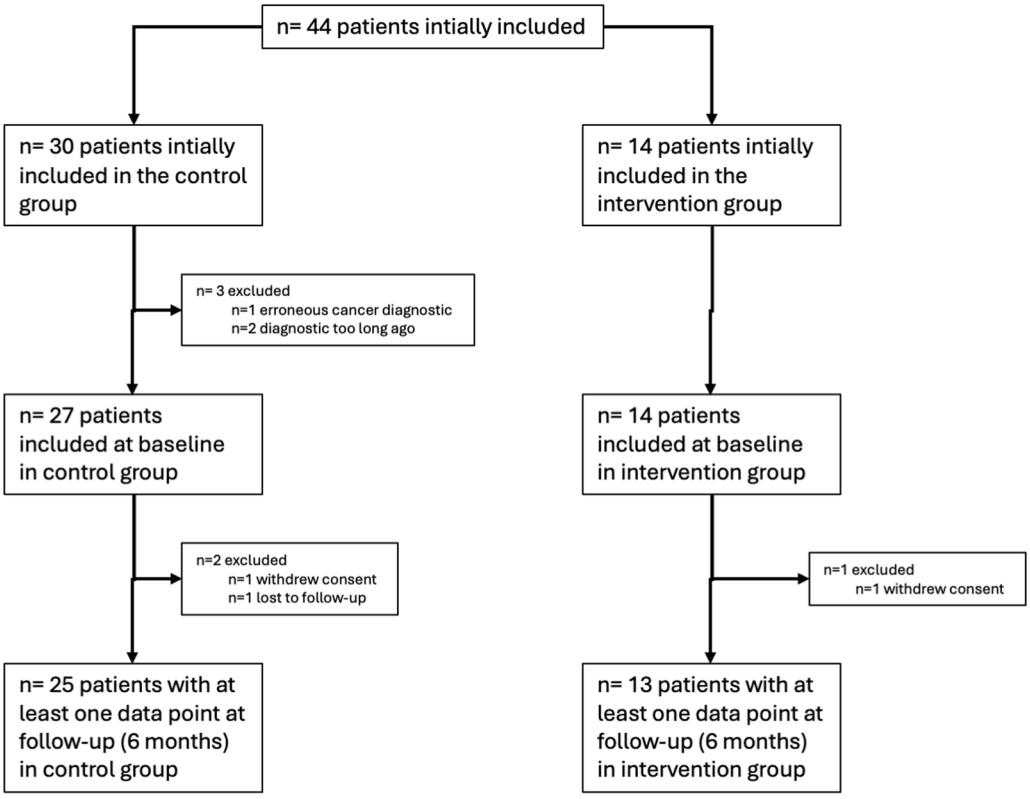


Fig. 1. Participant flow diagram.

proportion of retired participants. These differences may have influenced practical availability for an additional 90-minute off-site consultation and should therefore be considered when interpreting recruitment, satisfaction, and health-literacy outcomes. No formal between-group comparison was performed, as the study was not powered for such analyses.

3.2. Feasibility outcomes

During the intervention phase, in addition to those included, 49 patients were eligible and approached to participate yet did not provide consent to participate. Several reasons were voluntarily provided by these people such as not wanting to go to another location than the hospital for the intervention (n = 13), organisational issues with scheduling (n = 5), too many emotions at the time being (n = 3),

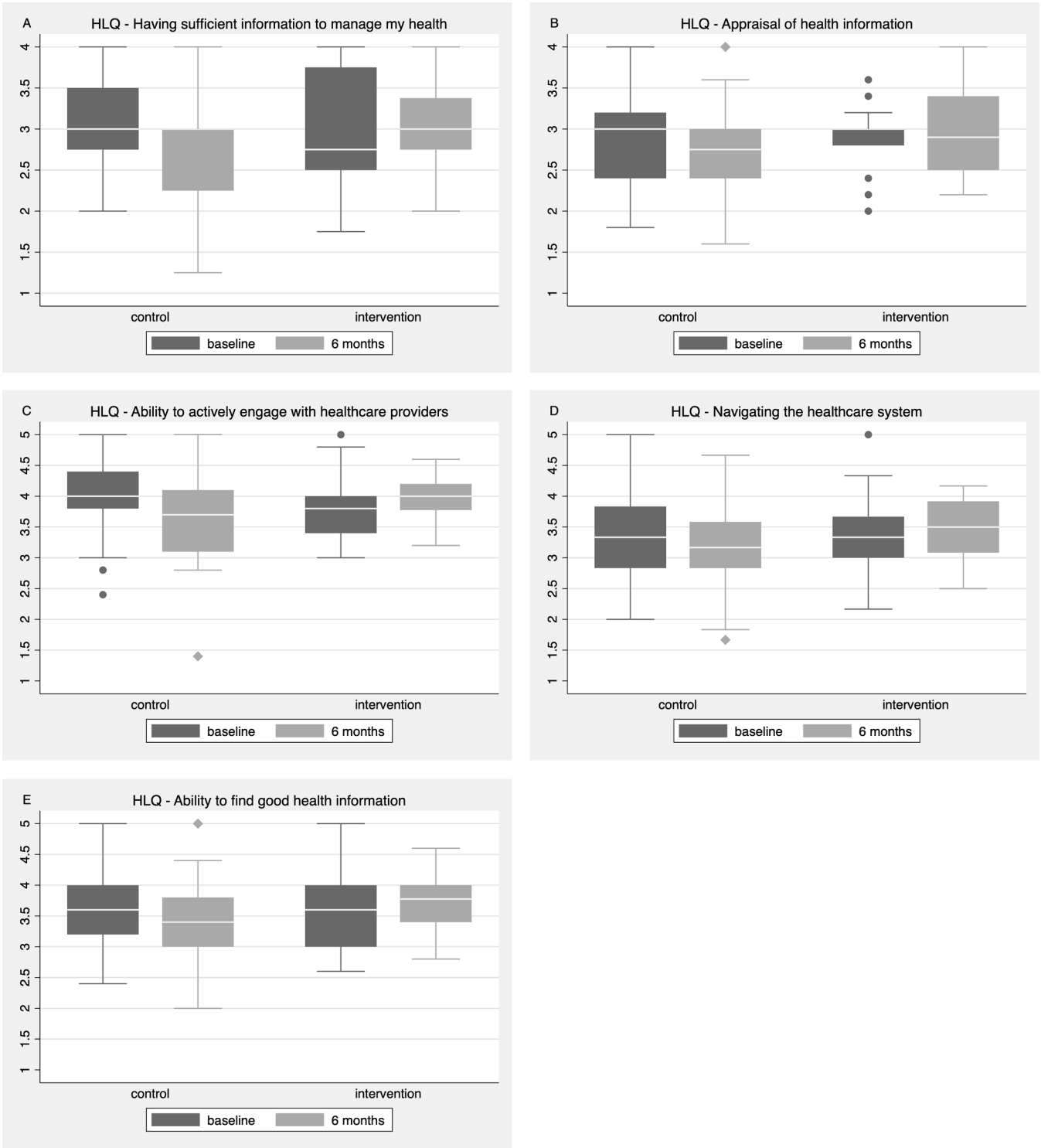


Fig. 2. Health Literacy Questionnaire results for five subscales (A-E) for both the control group and intervention group at baseline and 6-month follow-up. Analysis was descriptive given the study aims and the non-randomized design.

difficulties with autonomy or disease (n = 3), not interested (n = 2), too many medical appointments already (n = 2), or already feeling informed (n = 1).

Overall, the intervention recruited 14 participants over 6 months, demonstrating a low recruitment rate. Under the implementation conditions tested in this study, recruitment feasibility was therefore limited. This difficulty mainly concerned reach and uptake, rather than the ability to conduct the consultation among patients who consented. Reasons for refusal suggest that early timing after diagnosis, off-site delivery, and scheduling burden were central barriers.

3.3. Care pathways outcomes

Regarding health literacy, the HLQ has nine subscales that were analysed independently. Across the nine HLQ subscales, mean scores were generally stable over time in both groups. HLQ results are reported descriptively. Across subscales, changes were small and heterogeneous. Some domains, including access to information, engagement with providers, navigation, and finding good health information, showed descriptive trends in a favourable direction in the intervention group and modest declines in the control group. However, given the small sample size, group imbalance, and deliberate absence of inferential testing, these patterns should be interpreted as exploratory and hypothesis-generating.

Going into more detail, “feeling understood and supported by healthcare providers” showed minimal differences between groups and little change from baseline to follow-up. “Having sufficient information to manage my health” decreased in the control group and increased slightly in the intervention group (Fig. 2A). “Actively managing my health” increased modestly from baseline in the intervention group, with no material change in the control group. “Social support for health” was largely stable in the intervention group, with a slight decrease observed in the control group. “Appraisal of health information” displayed a mixed pattern without a consistent direction across groups (Fig. 2B). “Ability to actively engage with healthcare providers” decreased in the control group and increased slightly in the intervention group (Fig. 2C). “Navigating the healthcare system” showed a slight decrease in the control group and a slight increase in the intervention group (Fig. 2D). “Ability to find good health information” decreased in the control group and increased in the intervention group (Fig. 2E). Understanding health information well enough to know what to do changed little in either group.

3.4. CAT-related outcomes

Regarding attitudes towards CAT, as measured by CHBQ scores, there was no difference between groups at baseline or follow-up, and little intra-group evolution. The proportion with a positive attitude (CHBQ>40) was high at both time points (control 76.9% (20/26) at baseline and 78.3% (18/23) at 6 months; intervention 71.4% (10/14) and 75.0% (9/12)).

Regarding CAT use, it increased similarly in both groups between inclusion and follow-up. Data regarding satisfaction were heterogeneous, as the number of participants was too small to observe any meaningful changes. These results do not support any conclusion about the effect of Kit-CAT on CAT use, but they suggest that attitudes toward CAT were already favourable at baseline and remained broadly stable.

Regarding safety, no serious adverse events were reported. Adverse events were reported for 25 participants (16 in the control group and 9 in the intervention group), but none was considered related to Kit-CAT; they mainly reflected events related to cancer surgery, chemotherapy, radiotherapy, endocrine therapy, or intercurrent conditions.

3.5. Care satisfaction

Satisfaction outcomes are reported with available-case denominators because not all participants answered each item. Denominators are reported separately for each group and item. Satisfaction with their current care pathway was higher in the intervention group than in the control group: 78% (18/23) in the control group and 92% (11/12) in the intervention group reported being satisfied or very satisfied (control (n = 23): 13 satisfied, 5 very satisfied; intervention (n = 12): 8 satisfied, 3 very satisfied). Because denominators were small and differed across outcomes, these percentages should be interpreted cautiously.

Regarding their satisfaction with oncology care more specifically, both groups reported similar satisfaction levels, with the intervention group being slightly more favourable. One participant of the control group was slightly unsatisfied, and 14 were satisfied and 8 were very satisfied (n = 23). All participants (12/12) of the intervention group were either satisfied (6/12) or very satisfied (6/12).

Satisfaction with the intervention itself was overall high right after the intervention and slightly lower at follow-up. Some participants highlighted that they expected to receive more prescriptive advice (further highlighted in the qualitative study [24]). Nearly 60% (7/12) of respondents strongly agreed with recommending this intervention to future patients. This finding supports acceptability among participants who received the consultation, but not the feasibility of recruitment at scale.

4. Discussion

This pilot study developed a novel intervention aiding healthcare providers in discussing CAT with patients consulting for cancer. It then examined the feasibility and preliminary effects of this serious game to structure discussions about CAT in early breast cancer care. Because of the small sample size, group imbalance, non-randomized before-and-after design, and descriptive analysis, these findings should be interpreted as feasibility and hypothesis-generating data rather than evidence of effectiveness. Regarding our primary outcome, recruitment was significantly lower than expected due to several logistical reasons, as discussed in the study limitations. From the results, three signals emerge. First, the intervention was acceptable to participating patients and trained primary care physicians, with high immediate satisfaction and a sustained willingness to recommend the consultation at 6 months. Second, across several HLQ subscales, small descriptive trends were observed in the intervention group - contrasting with modest declines in the control group - but these cannot be interpreted causally. Third, no serious adverse events were reported, and recorded adverse events were not considered related to Kit-CAT.

4.1. Interpretation and relation to prior work

Consistent with observational and review evidence showing that proactively discussing CAT in a patient-centered, communicative style is linked to more patient-centered conversations, higher satisfaction with visits, and a stronger therapeutic relationship [13,16], and with communication frameworks promoting non-judgmental, evidence--based dialogue to reduce distress, set realistic expectations, and support informed choices [17,20], our findings suggest that a structured, ethically grounded conversation aid can assist women in aligning expectations with evidence limits, reducing anxiety, and better navigating services during early breast cancer care [10,14,15]. However, the present study does not demonstrate that Kit-CAT improves these outcomes. Rather, it suggests that these mechanisms are plausible and should be tested in future adequately powered evaluations.

In line with literature linking health literacy to cancer outcomes [28, 29], and consistent with the nine-domain framework of the HLQ underpinning our measurement strategy [25], the observed trends - although small and heterogeneous - are coherent with a mechanism whereby providing information, clarifying benefits and risks (including interactions), and modelling critical appraisal contribute to better engagement with providers and systems [17,20,21]. The HLQ findings

should not be interpreted as evidence that Kit-CAT improves health literacy. Rather, they suggest that selected HLQ domains may be relevant candidate outcomes for a future adequately powered evaluation. Alternative explanations, including selection bias, period effects, baseline differences, regression to the mean, and random variation, cannot be excluded. Given the high baseline prevalence of positive attitudes towards CAT in oncology, maintaining realism about potential benefits while safeguarding against harm remains a central ethical and educational challenge.

4.2. Strengths and limitations

As a pilot study, our estimates are not designed for hypothesis testing. The value of this study lies in demonstrating the acceptability and identifying implementation barriers (timing, venue, and duration), which will inform a pragmatic, multicenter evaluation of effectiveness and scale-up. Strengths include a theory-informed, iteratively prototyped intervention delivered in real-world settings, as well as multidimensional outcomes collected longitudinally.

However, several limitations warrant emphasis. First, the study was small and monocentric, using a quasi-experimental before-and-after design that is vulnerable to several biases. Temporal or period effects cannot be excluded: the control and intervention groups were recruited during different calendar periods, and contextual factors such as clinical workload, staff availability, patient flow, and seasonal organization may have differed between periods. The control recruitment period was not continuous, and recruitment procedures were refined between recruitment phases, which may have influenced screening and inclusion processes. Selection bias is also likely in the intervention period, because patients who accepted an additional 90-minute off-site consultation may have differed from those who declined in terms of availability, emotional readiness, interest in CAT, socioeconomic constraints, or perceived information needs. For these reasons, observed between-group differences cannot be interpreted causally and should be regarded only as exploratory signals to inform future research.

Recruitment faced considerable obstacles in October, coinciding with France National Breast Cancer Awareness Month. Consequently, hospital teams were primarily focused on awareness efforts rather than recruitment activities. Significantly, recruitment during the intervention period was also constrained by the very early timing after diagnosis, a phase marked by high emotional load and logistical demands (multiple concurrent appointments, travel to a different site). Suggesting the intervention early on in the care pathway allowed it to arrive before any early contact with CAT as a result of the cancer diagnosis. However, in hindsight, it may have led to the intervention being too early in the care pathway, contributing to refusals. Similarly, the intervention was suggested outside of the hospital to allow for a more neutral location for patients to discuss CAT without fear of judgment. In hindsight, the added logistical constraints discouraged some patients from enrolling. The resulting lower acceptance rate should therefore be interpreted primarily as a key feasibility discovery of an implementation barrier - related to timing, venue, and care-trajectory pressures - rather than as a failure of the intervention itself.

The small sample size and imbalance between groups further limit interpretation. In particular, the intervention group included only 13 participants with at least one 6-month follow-up data point, meaning that proportions and satisfaction estimates are highly unstable and sensitive to the response of one or two participants. Similarly, descriptive changes in HLQ subscales may reflect baseline differences, regression to the mean, period effects, selection processes, or random variation rather than intervention-related changes. These results should therefore be used for hypothesis generation and planning of future studies, not as evidence of effectiveness.

Intervention fidelity was supported by facilitator training and a structured guide, but it was not formally assessed. Favoring external validity, consultations were not audio-recorded or independently coded, and we did not quantify adherence to each component of the intervention. Consequently, variability in facilitation style, depth of discussion, and time spent on each module may have occurred.

4.3. Implications for practice and research

The findings support the feasibility and perceived value of structured CAT conversations in oncology, while underscoring the need for scalable delivery models. More precisely, the findings support the acceptability and deliverability of Kit-CAT among consenting participants, but they do not demonstrate recruitment feasibility under the tested implementation conditions. However, a 90-minute consultation would not be feasible in the current healthcare system due to existing time constraints on medical personnel. Therefore, future studies might investigate (i) alternative staffing solutions (e.g., nurse- or supportive-care-led consultations), (ii) flexible timing for patients (earlier for information needs vs later for readiness; or stepwise offers), (iii) hybrid/telehealth and digital versions of the game to reduce travel and scheduling burden, and (iv) group formats where appropriate.

Recruitment difficulties appear to be related primarily to the implementation model rather than to rejection of the intervention concept itself. The most frequent reasons for refusal concerned the off-site location, scheduling constraints, emotional burden shortly after diagnosis, autonomy or disease-related difficulties, and the already high number of medical appointments. These barriers suggest that the tested format - a 90-minute consultation delivered outside the hospital within two weeks of inclusion - may have been insufficiently aligned with patients' readiness and logistical constraints. Future versions should therefore consider shorter or modular consultations, hospital-based or telehealth delivery, flexible timing, integration into supportive-care pathways, and delivery by trained nurses or supportive-care professionals.

A pragmatic multicentric trial with mixed-methods evaluation and longer follow-up is warranted to assess effects on health literacy, communication quality, decisional conflict, care experience, and safety (including drug-herb interactions), while prospectively capturing implementation determinants (reach, fidelity, dose) to inform adoption at scale. Such a trial should be preceded by optimization of the implementation format and should include predefined feasibility progression criteria. A key implication is that future studies should not simply reproduce the same format at larger scale. They should first adapt the implementation model to improve reach while preserving the core components of Kit-CAT: non-judgmental dialogue, evidence-informed appraisal, clarification of expectations, and preparation for discussion with the oncology team.

5. Conclusion

This pilot study developed and implemented Kit-CAT, a serious-game-supported consultation designed to facilitate structured, non-judgmental, and evidence-informed discussions about CAT in early breast cancer care. Among patients who received the intervention, acceptability was high; no serious adverse events were reported, and recorded adverse events were not considered related to Kit-CAT. Descriptive trends suggested possible favourable changes in selected health-literacy domains, but these findings remain exploratory because of the small sample size, group imbalance, and non-randomized before-and-after design. The main feasibility challenge concerned recruitment, particularly the early timing after diagnosis, the 90-minute duration, and the off-site delivery of the consultation. Overall, Kit-CAT appears conceptually acceptable and deliverable among consenting participants, but further adaptation of its implementation format is required before evaluating its effectiveness in a larger multicentric study.

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Clinical trial registration and ethics approval

This study was registered under ID/RCB 2021-A03027–34 and received ethical approval from a national ethics committee, with the number 21.04547.000055. It was also registered on ClinicalTrials.gov under the number NCT05388734.

CRediT authorship contribution statement

Nicolas Pinsault: Writing – original draft, Supervision, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Pauline Girard:** Writing – review & editing, Investigation, Conceptualization. **Richard Monvoisin:** Writing – review & editing, Project administration, Conceptualization. **Claire Eychenne:** Writing – review & editing, Project administration, Methodology. **Céline Genty-Vermorel:** Validation, Formal analysis, Data curation. **Leo Druart:** Writing – original draft, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare having no conflicts of interest in relation to this project and article.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.ctarc.2026.101272.

References

[1] B. Lapôtre-Ledoux, L. Remontet, Z. Uhry, E. Dantony, P. Grosclaude, F. Molinié, et al., Incidence des principaux cancers en France métropolitaine en 2023 et tendances depuis 1990, Bull. Épidémiol. Hebd. (12–13) (2023) 188–204. Available from, http://beh.santepubliquefrance.fr/beh/2023/12-13/2023_12-13_1.html.
[2] A. Sparber, J.C. Wootton, Surveys of complementary and alternative medicine: part II. Use of alternative and complementary cancer therapies, J. Altern. Complement. Med. 7 (2001) 281–287, https://doi.org/10.1089/107555301300328179.
[3] M.R. Keene, I.M. Heslop, S.S. Sabesan, B.D. Glass, Complementary and alternative medicine use in cancer: a systematic review, Complement. Ther. Clin. Pract. 35 (2019) 33–47, https://doi.org/10.1016/j.ctcp.2019.01.004.
[4] M. Horneber, G. Bueschel, G. Dennert, D. Less, E. Ritter, M. Zwahlen, How many cancer patients use complementary and alternative medicine: a systematic review and metaanalysis, Integr. Cancer Ther. 11 (2012) 187–203.
[5] E. Ernst, Complementary and alternative medicine (CAM) and cancer: the kind face of complementary medicine, Int. J. Surg. 7 (2009) 499–500, https://doi.org/10.1016/j.ijsu.2009.08.005.
[6] W.A. Weiger, M. Smith, H. Boon, M.A. Richardson, T.J. Kaptchuk, D.M. Eisenberg, Advising patients who seek complementary and alternative medical therapies for cancer, Ann. Intern. Med. 137 (2002) 889–903, https://doi.org/10.7326/0003-4819-137-11-200212030-00010.
[7] M. Tascilar, F.A. de Jong, J. Verweij, R.H.J. Mathijssen, Complementary and alternative medicine during cancer treatment: beyond innocence, Oncologist 11 (2006) 732–741, https://doi.org/10.1634/theoncologist.11-7-732.
[8] R. Firkins, H. Eisfeld, C. Keinki, J. Buentzel, A. Hochhaus, T. Schmidt, et al., The use of complementary and alternative medicine by patients in routine care and the risk of interactions, J. Cancer Res. Clin. Oncol. 144 (2018) 551–557, https://doi.org/10.1007/s00432-018-2587-7.
[9] J. Hubner, V. Hanf, Commonly used methods of complementary medicine in the treatment of breast cancer, Breast. Care 8 (2013) 341–347, https://doi.org/10.1159/000355705.
[10] I. Juraskova, L. Hegedus, P. Butow, A. Smith, P. Schofield, Discussing complementary therapy use with early-stage breast cancer patients: exploring the communication gap, Integr. Cancer Ther. 9 (2010) 168–176, https://doi.org/10.1177/1534735410365712.
[11] L.G. Balneaves, T.L.O. Truant, M. Kelly, M.J. Verhoef, B.J. Davison, Bridging the gap: decision-making processes of women with breast cancer using complementary and alternative medicine (CAM), Support. Care Cancer 15 (2007) 973–983, https://doi.org/10.1007/s00520-007-0282-3.
[12] G. Yang, H. Zhang, Z. Gan, Y. Fan, W. Gu, C. Ling, Discrepant views of oncologists and cancer patients on complementary and alternative medicine in a chinese general hospital, Integr. Cancer Ther. 17 (2018) 451–457, https://doi.org/10.1177/1534735417725579.
[13] E.L. Davis, B. Oh, P.N. Butow, B.A. Mullan, S. Clarke, Cancer patient disclosure and patient-doctor communication of complementary and alternative medicine use: a systematic review, Oncologist 17 (2012) 1475–1481, https://doi.org/10.1634/theoncologist.2012-0223.
[14] J. Ge, J. Fishman, N. Vapiwala, S.Q. Li, K. Desai, S.X. Xie, et al., Patient-physician communication about complementary and alternative medicine in a radiation oncology setting, Int. J. Radiat. Oncol. Biol. Phys. 85 (2013) e1–e6, https://doi.org/10.1016/j.ijrobp.2012.08.018.
[15] M.J. Verhoef, H.S. Boon, S.A. Page, Talking to cancer patients about complementary therapies: is it the physician's responsibility? Curr. Oncol. 15 (2008) s88–s93, https://doi.org/10.3747/co.v15i0.279, 15 Suppl 2.
[16] D.L. Roter, K.J. Yost, T. O'Byrne, M. Branda, A. Leppin, B. Kimball, et al., Communication predictors and consequences of complementary and Alternative Medicine (CAM) discussions in oncology visits, Patient. Educ. Couns. 99 (2016) 1519–1525, https://doi.org/10.1016/j.pec.2016.06.002.
[17] M. Frenkel, E. Ben-Arye, L. Cohen, Communication in cancer care: discussing complementary and alternative medicine, Integr. Cancer Ther. 9 (2010) 177–185, https://doi.org/10.1177/1534735410363706.
[18] S.H. Kim, D.W. Shin, Y.-S. Nam, S.Y. Kim, H.-K. Yang, B.L. Cho, et al., Expected and perceived efficacy of complementary and alternative medicine: a comparison views of patients with cancer and oncologists, Complement. Ther. Med. 28 (2016) 29–36, https://doi.org/10.1016/j.ctim.2016.07.005.
[19] R.L. Street, Jr, G. Makoul, N.K. Arora, R.M Epstein, Pathways linking clinician-patient communication to health outcomes, Patient. Educ. Couns. 74 (2009) 295–301, https://doi.org/10.1016/j.pec.2008.11.015.
[20] M. Frenkel, L. Cohen, Effective communication about the use of complementary and integrative medicine in cancer care, J. Altern. Complement. Med. 20 (2014) 12–18, https://doi.org/10.1089/acm.2012.0533.
[21] N. Sharifzadeh, H. Kharrazi, E. Nazari, H. Tabesh, M. Edalati Khodabandeh, S. Heidari, et al., Health education serious games targeting Health care providers, patients, and public Health users: scoping review, JMIR. Serious. Games. 8 (2020) e13459, https://doi.org/10.2196/13459.
[22] T.C. Hoffmann, P.P. Glasziou, I. Boutron, R. Milne, R. Perera, D. Moher, et al., Better reporting of interventions: template for intervention description and replication (TIDieR) checklist and guide, BMJ 348 (2014) g1687, https://doi.org/10.1136/bmj.g1687.
[23] D.C. Des Jarlais, C. Lyles, N. Crepaz, T.R.E.N.D. Group, Improving the reporting quality of nonrandomized evaluations of behavioral and public health interventions: the TREND statement, Am. J. Public Health 94 (2004) 361–366, https://doi.org/10.2105/ajph.94.3.361.
[24] G. Coisnon, C. Truong, N. Pinsault, R. Monvoisin, P. Girard, L. Druart, « Se poser, trier, et garder sa boussole ». Vecu d'une consultation d'education en sante sur les therapies complementaires, Bull. Cancer 112 (2025) 1406–1415, https://doi.org/10.1016/j.bulcan.2025.06.012.
[25] R.H. Osborne, R.W. Batterham, G.R. Elsworth, M. Hawkins, R. Buchbinder, The grounded psychometric development and initial validation of the Health Literacy Questionnaire (HLQ), BMC. Public Health 13 (2013) 658, https://doi.org/10.1186/1471-2458-13-658.
[26] D. Lie, J. Boker, Development and validation of the CAM Health belief Questionnaire (CHBQ) and CAM use and attitudes amongst medical students, BMC. Med. Educ. 4 (2) (2004), https://doi.org/10.1186/1472-6920-4-2.
[27] A. Guillaud, N. Darbois, C. Riboud, M. Dolgopoloff, B. Allenet, N. Pinsault, Development, validation and reliability of the CAM practitioner use questionnaire (CAMP-Q)], Sante Publique Vol. 31 (2019) 817–826, https://doi.org/10.3917/spub.196.0817.
[28] C.E. Holden, S. Wheelwright, A. Harle, R. Wagland, The role of health literacy in cancer care: a mixed studies systematic review, PLoS. One 16 (2021) e0259815, https://doi.org/10.1371/journal.pone.0259815.
[29] B. Al Hussein Al Awamlh, K.A. Moses, J. Whitman, T. Stewart, S. Kripalani, K. Idrees, Health literacy and all-cause mortality among cancer patients, Cancer 131 (2025) e35794, https://doi.org/10.1002/cncr.35794.